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Next item →

1. Execute the Coursera Lab Jupyter notebook for this lesson, which implements both the EKF and CKF.

1 / 1 point

Which estimator produces lower root-mean-squared errors (RMSE) and which produces more reliable confidence bounds?

- ☐ The EKF produces lower RMSE and more reliable confidence bounds.
- ☐ The EKF produces lower RMSE and the CKF produces more reliable confidence bounds.
- ☒ The CKF produces lower RMSE and more reliable confidence bounds.
- ☐ The CKF produces lower RMSE and the EKF produces more reliable confidence bounds.

✔ Correct
Yes, this is correct.

2. For the same Coursera Lab Jupyter notebook, what is the rms error for the CKF for the **position** state? Enter your answer with four digits to the right of the decimal point. Note that the code is originally configured to compute the rms error for all the states (position and velocity) so you will need to make minor modifications to compute the desired result.

1 / 1 point

0.0031

✔ Correct
Yes. That is correct.

3. For the same Coursera Lab Jupyter notebook, what is the percentage of time that the CKF's **position** estimate is outside of its estimation confidence bounds? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

0.8

✔ Correct
Yes, that is correct.